

Question Paper

Exam Date & Time: 21-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Social Network Analysis [NPT_4146]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

1) Consider the following code snippet and answer the questions below: (5)

A)

```
import networkx as nx
G = nx.Graph()
G.add_nodes_from(["Rahul", "Sneha", "Amit", "Priya", "Karan"])
G.add_edges_from([("Rahul", "Sneha"), ("Rahul", "Amit"), ("Sneha", "Priya"), ("Amit", "Karan")])
G.remove_edge("Rahul", "Amit")
G.remove_node("Priya")
has_edge = G.has_edge("Rahul", "Amit")
neighbors_of_rahul = list(G.neighbors("Rahul"))
```

Apply and interpret how the application of `remove_edge()`, `remove_node()`, `has_edge()`, and `neighbors()` functions modifies the underlying social network structure. Further, evaluate the resulting network connectivity and construct an extended version of the graph by adding a new connection between Sneha and Karan. Explain how this modification influences the relationships among the remaining individuals.

Course Outcome: "CO1" / Cognitive Level: "Application"

B) Apply graph metrics such as **betweenness centrality**, **closeness centrality**, and **modularity** in **Gephi** to analyze and interpret the key structural properties of a network. Examine how these metrics reveal influential nodes and community structures, and illustrate their computation and visualization within the software. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

C) Apply the concept of **betweenness centrality** to analyze and interpret the role of intermediary nodes in facilitating information flow within large, real-world networks such as social media platforms. Examine how variations in betweenness values reflect node influence and discuss the implications of these findings for understanding key connectors in the network. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

2) Consider a node v in a social network. This node has **4 neighbors**: A, B, C, and D. The connections between these neighbors are as follows: (5)

- A)
- **A is connected to B and C.**
 - **B is connected to C.**
 - **C is connected to D.**

Analyze the local clustering coefficient of node v in the given social network with neighbors **A, B, C,**

and **D**, considering the specified interconnections among the neighbors, and interpret what the resulting value reveals about the cohesiveness of node **v**'s neighborhood.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) The following table represents a **social network of six individuals** with their **attributes and friendships**: (3)

Person	Gender	Occupation	Friends
Alice	Male	Teacher	Bob, Frank
Bob	Male	Engineer	Alice, Carol
Carol	Female	Teacher	Bob, Eve
Eve	Female	Engineer	Carol, Grace
Grace	Female	Doctor	Eve, Frank
Frank	Male	Engineer	Grace, Alice

Examine the given network to identify **two instances of homophily**-cases where friendships occur between individuals sharing the same **gender** or **occupation**. Discuss what these patterns indicate about the **tendency of people to form connections based on shared attributes** in social networks.

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) **Apply** the concept of **social influence** in network analysis to determine who **Alice** with 200 connections or **Bob** with 50 connections-would likely have a greater impact when sharing the same post, and justify your reasoning based on network connectivity principles. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) **Apply** the concept of **network isolation** to explain how **spatial segregation** leads to the formation of **isolated clusters** within a network, and interpret the implications this has for **information flow**, **access to diverse perspectives**, and **social mobility** in society (5)

A)

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) The following table represents the **relationships** among five individuals in a small social group: (3)

Relationship	Type of Relationship
A - B	Positive (+)
A - C	Negative (-)
B - C	Positive (+)
B - D	Negative (-)
C - E	Positive (+)
D - E	Negative (-)

Examine the network using the concept of balance theory by identifying the total number of positive and negative relationships and determining which single relationship adjustment would achieve balance.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) Utilize the concept of balance theory to explain how changing one negative relationship to a positive one can transform an unstable triad into a stable configuration in a social network. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) **Examine** the **limitations** of using **power-law distributions** to model real-world social networks. (5)
Identify key **criticisms** related to how well the power-law model fits empirical network data, and analyze how researchers can **address these challenges** when interpreting or validating social network structures.
- A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Analyze how the "**richer gets richer**" effect (preferential attachment) influences **the degree distribution** of a social network and results in the formation of **highly influential nodes**. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Apply the concept of the long-tail phenomenon in a degree distribution within the context of Social Network Analysis, and explain what it reveals about the presence of a few highly connected nodes and many sparsely connected nodes in the network. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Utilize the **mathematical foundations** of the **small-world model** and explain **six degrees of separation, viral marketing, information diffusion in real-world applications**. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) **Apply** the **cascading model** to explain the process of **information diffusion**, highlighting how information spreads progressively from **one user to others** within a social network. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Utilize the concept of short path lengths to examine how information or viral content quickly propagates through a social network. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

NPT_4146:Social Network Analysis

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

NPT_4146:Social Network Analysis

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	11	34(68%)
4	Analysis	4	16(32%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)